

MFE5200 – Computational Methods in Financial Engineering
Summer Term, 2025-26
Assignment 3

Problem 1

Let $X_1(t)$ and $X_2(t)$ be two stochastic processes. The SDEs of $X_1(t)$ and $X_2(t)$ are given by

$$\begin{aligned}dX_1(t) &= a_1 X_1(t) dt + \sigma_1 X_1(t) dW_1(t), \\dX_2(t) &= a_2 X_2(t) dt + \sigma_2 X_2(t) dW_2(t),\end{aligned}$$

where a_1, a_2, σ_1 and σ_2 are all constants, $W_1(t)$ and $W_2(t)$ are two correlated Brownian motions and also suppose that $dW_1(t)dW_2(t) = \rho dt$.

Let $(R_1, V_1), (R_2, V_2), \dots, (R_m, V_m)$ be the i.i.d. samples from $N(\mathbf{0}, I_2)$ where $\mathbf{0} \in \mathbb{R}^2$ and I_2 is a 2×2 identity matrix.

On the time grid $0 = t_0 < t_1 < \dots < t_m$, write down the Euler Scheme for the stochastic differential equations of $X_1(t)$ and $X_2(t)$ in terms of $(R_1, V_1), (R_2, V_2), \dots, (R_m, V_m)$.

Problem 2

Let $\{Z_1, Z_2, \dots, Z_m\}$ be i.i.d. standard normal random variables.

On the time grid $0 = t_0 < t_1 < \dots < t_m$, write down the Euler Scheme and Milstein Scheme for the following stochastic differential equation in terms of $Z_1, Z_2, \dots, Z_m$

$$dX(t) = rX(t)dt + \sqrt{\sigma_0^2 + \sigma_1^2 X^2(t)} dW(t),$$

where $W(t)$ is a standard Brownian motion and all the constants involved are assumed to be strictly positive.

Problem 3

Find the Lamperti transform of the following stochastic differential equations. All the constants involved are assumed to be strictly positive.

(a) $dX(t) = a(b - X(t))dt + \sigma\sqrt{X(t)}dW(t).$

(b) $dX(t) = (r_0 + r_1X(t))dt + \sigma X(t)dW(t).$

Problem 4

Assume that the stock price is a geometric Brownian motion under the risk-neutral measure:

$$dS(t) = rS(t)dt + \sigma S(t)dW(t),$$

where r is the continuously compounded risk-free annual interest rate, σ is the volatility of the stock price and $W(t)$ is a standard Brownian motion under the risk-neutral measure.

Using the method of pathwise differentiation, the vega of a standard European call option with maturity T and strike price K at time 0 is found to be in the form of

$$E\left[\mathbf{1}_{\{S(T)>K\}}e^{-rT}f(S(0),S(T))\right],$$

where $E[\cdot]$ is the expectation under the risk-neutral measure.

Find $f(S(0), S(T))$.

Problem 5

Assume that the stock price is a geometric Brownian motion under the risk-neutral measure:

$$dS(t) = rS(t)dt + \sigma S(t)dW(t),$$

where r is the continuously compounded risk-free annual interest rate, σ is the volatility of the stock price and $W(t)$ is a standard Brownian motion under the risk-neutral measure.

Consider an arithmetic average option with strike price K and the payoff at the expiration T of

$$(\bar{S}_A - K)^+, \quad \bar{S}_A = \frac{1}{m} \sum_{i=1}^m S(t_i),$$

where $t_i = ih$ for $i = 1, \dots, m$ and $h = T/m$.

Use the likelihood ratio method, write down an estimate of the vega of that option at time 0 in terms of $Z_1, Z_2, \dots, Z_m$ where $Z_1, Z_2, \dots, Z_m$ are i.i.d. standard normal random variables.

Problem 6

For a 1.5-year American put option on a stock, you are given:

- i. The strike price is \$20.
- ii. The American option can only be exercised at 0.5 year, 1 year and 1.5 year.
- iii. The continuously compounded risk-free interest rate is 6%.
- iv. Under the risk-neutral measure, six sample paths for the price of the stock at time t are

Path	$t = 0.5$	$t = 1$	$t = 1.5$
1	21.74	22.82	17.16
2	17.98	15.84	25.89
3	28.07	38.63	37.94
4	10.38	13.26	10.67
5	21.98	14.68	11.05
6	18.88	22.78	34.57

Using the least squares approach by assuming that the expected payoff from continuation at time $t_i = 0.5i$, $i = 1, 2$, is modelled by a **linear function** of $S(t_i)$, calculate the value of the American option at time 0.

Note: The video about how to use the scientific calculator to do the linear regression has been uploaded to BB. The video only shows the use of a widely used model calculator to do the linear regression. If your model is different from it, you have to consult the user manual by yourself.

Problem 7

Consider a portfolio which consists of

Call on asset S_1	Asset S_1	Put on asset S_2	Asset S_2
Long 2 units	Long 5 units	Short 3 units	Short 4 units

You are given that

- The delta and gamma of the options are given by

	Call on asset S_1	Put on asset S_2
Delta	0.56	-0.27
Gamma	0.08	0.07

- The distribution of $\Delta S = (\Delta S_1, \Delta S_2)^T$ is given by

$$\Delta S \sim N(\mu, \Sigma),$$

where

$$\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} 0.5 & 0.18 \\ 0.18 & 0.3 \end{pmatrix}.$$

Let $V(t, S_1, S_2)$ be the value of the portfolio at time t . Assume that Δt is small and negligible.

- Write down the delta-gamma approximation Q for the portfolio loss $L = -\Delta V$ in terms of ΔS .
- Diagonalize Q in (a) and express it in terms of $Z = (Z_1, Z_2)^T$ where Z_1 and Z_2 are two independent standard normal random variables.

Note:

Consider a real 2×2 symmetric matrix

$$A = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \quad \text{where } c \neq 0.$$

The eigenvalues and the corresponding orthonormal eigenvectors are given by

$$\lambda_1 = \frac{1}{2}(a+b+D), \quad v_1 = \frac{1}{f_1}(2c, b-a+D)^T;$$

$$\lambda_2 = \frac{1}{2}(a+b-D), \quad v_2 = \frac{1}{f_2}(2c, b-a-D)^T,$$

where $D = \sqrt{(a-b)^2 + 4c^2}$, $f_1 = \sqrt{4c^2 + (b-a+D)^2}$ and $f_2 = \sqrt{4c^2 + (b-a-D)^2}$.

Problem 8

Define a family of probability measures P_θ through the likelihood ratio (9.10) on P.9 of the lecture notes.

Let Z be the standard multivariate normal under P .

Show that the distribution of Z under P_θ is a multivariate normal

$$Z \sim N(\mu(\theta), \Sigma(\theta)),$$

where $\Sigma(\theta)$ is a diagonal matrix with diagonal entries $\sigma_j^2(\theta)$,

$$\mu_j(\theta) = \frac{\theta b_j}{1 - 2\lambda_j \theta}, \quad \sigma_j^2(\theta) = \frac{1}{1 - 2\lambda_j \theta}.$$

Problem 9

Suppose that $X_1, X_2, \dots, X_n$ are independent observations that follow $N(\mu, \sigma^2)$, where μ is a known quantity.

(a) Show that the likelihood function $L(\sigma^2)$ satisfies

$$L(\sigma^2) \propto (\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \right].$$

(b) Suppose further that $\sigma^2 \sim IG(\alpha, \beta)$. What is the conditional distribution of $\sigma^2 | X_1, X_2, \dots, X_n$?

Problem 10

Consider the following transition matrix of a Markov chain:

	1	2	3	4
1	1/6	0	1/2	1/3
2	0	1/3	1/3	1/3
3	0	1/2	0	1/2
4	1/4	1/4	1/4	1/4

Use the Metropolis-Hastings algorithm to construct a Markov chain whose limiting distribution is $(1/6, 1/6, 1/3, 1/3)$ based on the aforementioned matrix.

Problem 11

In Example 12.6 of Chapter 9, it is required to sample from a standard normal probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad x \in (-\infty, \infty)$$

using the Metropolis-Hastings algorithm from a Markov chain $\{X_n, n = 0, 1, \dots\}$ with a transition probability

$$q(x_n | x_{n-1}) = \frac{1}{2} e^{-|x_n|}, \quad x_n \in (-\infty, \infty) \text{ and } n = 0, 1, \dots$$

Using the $U(0, 1)$ random numbers U , V and W below to sample four candidates from the Metropolis-Hastings algorithm. Indicate clearly which candidate variates are accepted (state of Markov chain changes) and which are rejected (state unchanged). Start with $X_0 = 0$.

(U, V) : for candidate variate	(0.18, 0.87)	(0.02, 0.42)	(0.31, 0.08)	(0.12, 0.79)
W : for acceptance test	0.55	0.03	0.18	0.43

where U is used to generate the sample from the exponential distribution and V is used to determine the sign of the generated exponential distributed sample.

End

NORMAL DISTRIBUTION TABLE

Entries represent the area under the standardized normal distribution from $-\infty$ to z , $\Pr(Z < z)$

The value of z to the first decimal is given in the left column. The second decimal place is given in the top row.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Values of z for selected values of $\Pr(Z < z)$							
z	0.842	1.036	1.282	1.645	1.960	2.326	2.576
$\Pr(Z < z)$	0.800	0.850	0.900	0.950	0.975	0.990	0.995